uBLOX MAX-M10S Breakout for Active Antennas

Description

The Uputronics u-blox MAX-M10S Breakout for active antennas. It permits the connection of the Ublox MAX-M10S GPS module to either 3 or 5V a microcontroller depending on board version.

Pin Configurations



SCL	I²C Clock Line.
SDA	I²C Data Line.
TX	Serial TX Line. Connect to microcontroller RX line.
RX	Serial RX. Connect to microcontroller TX line.
EN	5V Board Only Enables the level convertor. On these boards it is hard tied to VCC. See 5V Board Notes.
TP	Time Pulse Output. When Locked the GPS will output a 1hz pulse to this pin.
VCC	Supply Pin
GND	GND

Board Specifications

Weight	5g max
Battery Backup	0.2F Elna SuperCap
Connector Pitch	2.54mm
Default baud rate	9600bps
Power Usage	Acquire 25mA / Tracking 21mA / Cyclic PSM Mode 6mA

Board Absolute Maximums

3V Board VCC Max Voltage	3.6V
5V Board VCC Max Voltage	30V (Max logic level 5.5V)
Operating Temperature	-40°C to +85°C

GPS Performance

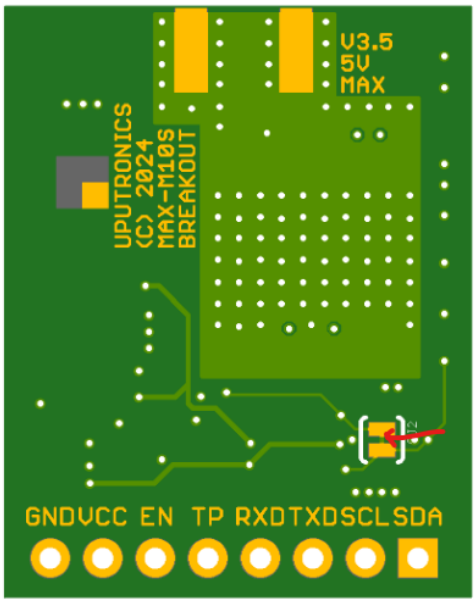
Data obtained from u-blox M10S GPS/GNSS Module Data sheet:

https://www.u-blox.com/sites/default/files/MAX-M8-FW3_DataSheet_%28UBX-15031506%29.pdf

Parameter	Specification		
Antenna Connector	SMA female	Antenna bias voltage	2.7V
Receiver Type	u-blox M10 engine GPS L1 C/A, QZSS L1 C/A L1S, GLONASS L1OF BeiDou B1I/B1C, Galileo E1B/C SBAS L1 C/A: WAAS, EGNOS, MSAS, GAGAN		
Time To First Fix	Cold Start Warm Start Hot Start	28s GPS/GLONASS 1s 1s	
Horizontal Position accuracy	2.5m GPS		
Max Navigation Rate	25Hz (Single GNSS) 10Hz (4 concurrent)		
Frequency of Time Pulse	0.25Hz -> 10Mhz		Configurable.
Operational Limits	Dynamics Altitude Velocity	< 4G 50,000m / 164000 feet 500m/s / 1118 mph	**These figures assume Airborne < 4g dynamic model.**

5V Board Notes

The EN is tied up to VCC on the board so the level convertor is always enabled. Should you require to switch this via a microcontroller you can cut the PCB track on the rear of the board:



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